

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA51 **COURSE NAME:** Cryptography and Network Security

COURSE OUTCOME COVERED

CO4: *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO (BL3, BL6)*

Assessment Tool Weightage: 10%

QUESTIONS: 5

TOTAL MARKS: 25

Annexure A

Comparative Analysis

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A
Comparative Analysis

1. Compare Authentication Header (AH) and Encapsulating Security Payload (ESP) in IPSec with respect to security services, performance overhead, and suitability for different network scenarios. Analyze which is more appropriate for a VPN deployment.
2. Compare transport mode and tunnel mode in IPSec architecture. Evaluate their effectiveness in securing host-to-host versus network-to-network communication.
3. Perform a comparative analysis of Pretty Good Privacy (PGP) and S/MIME in terms of key management, trust models, scalability, and enterprise usability. Recommend the most suitable approach for a corporate email system.
4. Compare rule-based and machine learning-based email spam detection techniques. Analyze their effectiveness in handling evolving spam patterns and minimizing false positives.
5. Compare manual key management and automated key management (e.g., using IKE) in IPSec. Evaluate their impact on security, scalability, and operational complexity in large organizations.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

COMPARATIVE ANALYSIS – QUESTION 1

Title

Comparison of Authentication Header (AH) and Encapsulating Security Payload (ESP) in IPSec

1. Introduction

IPSec is a collection of protocols used to secure communication at the IP layer. It is widely used in Virtual Private Networks (VPNs) to protect data transmitted over untrusted networks such as the public Internet.

The two major IPSec security protocols are:

- **Authentication Header (AH)**
- **Encapsulating Security Payload (ESP)**

Both can provide integrity and authentication, but they differ significantly in their ability to provide confidentiality, their processing overhead, and their suitability for modern VPN deployments.

2. Authentication Header (AH)

Authentication Header provides authentication and integrity protection for IP packets.

The main security services provided by AH are:

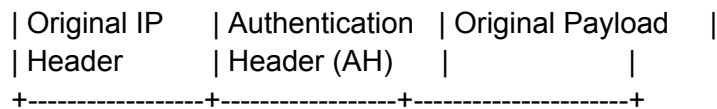
- Data integrity
- Data-origin authentication
- Anti-replay protection

AH does **not encrypt the packet payload**.

Therefore, the information contained in the payload remains readable to an observer who can capture the traffic.

Simplified AH Structure

+-----+-----+-----+



AH calculates an authentication value over protected portions of the packet.

If the packet is modified during transmission, authentication verification fails.

3. Encapsulating Security Payload (ESP)

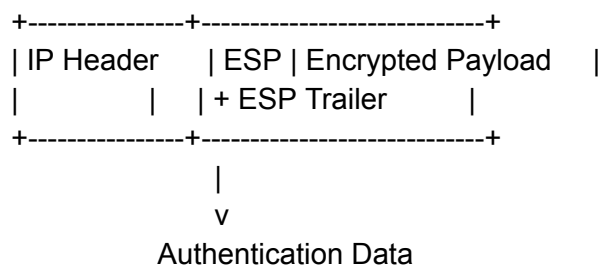
ESP is designed to provide a broader range of security services.

It can provide:

- Confidentiality
- Data integrity
- Data-origin authentication
- Anti-replay protection

ESP uses encryption to protect sensitive information.

Simplified ESP Structure



In modern deployments, ESP is commonly used with authenticated encryption or with separate integrity/authentication mechanisms as appropriate.

4. Security Services Comparison

Security Service	AH	ESP
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Confidentiality	No	Yes
Data integrity	Yes	Yes
Data-origin authentication	Yes	Yes
Anti-replay protection	Yes	Yes
Encryption	No	Yes
Suitable for sensitive data	Limited	Yes
Suitable for VPNs	Limited	Highly suitable

The major difference is **confidentiality**.

AH authenticates and protects integrity but does not hide the contents of the communication.

ESP can encrypt the protected data and therefore provides confidentiality.

5. Performance Overhead

AH

AH generally requires less cryptographic processing because it does not perform encryption.

```

Packet
|
v
Authentication
|
v
Send

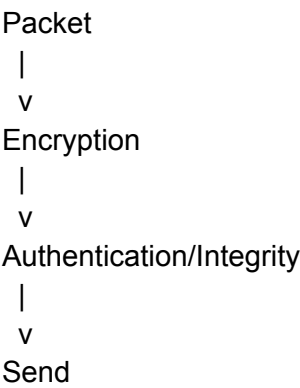
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Therefore, AH can have lower computational overhead in scenarios where confidentiality is not required.

However, AH can introduce networking complications, particularly with NAT, because AH authenticates parts of the IP header and NAT modifies address information.

ESP

ESP generally requires more processing because encryption is normally performed.



Therefore, ESP may have greater computational and bandwidth overhead.

However, modern processors and network devices often provide cryptographic acceleration, making ESP practical for high-speed VPNs.

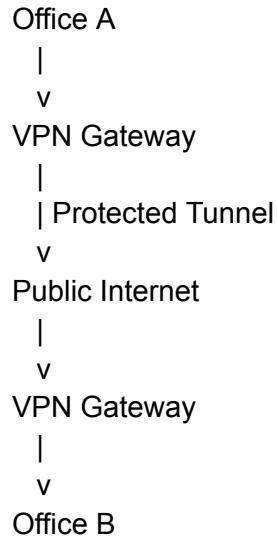
6. Comparison of Performance

Factor	AH	ESP
Encryption processing	None	Required when confidentiality is used
Authentication processing	Yes	Yes, when configured
CPU requirement	Generally lower	Generally higher
Packet overhead	Lower	Higher
Confidentiality	Not available	Available
NAT compatibility	Poorer	Better with NAT traversal
Modern VPN suitability	Limited	High

7. Suitability for Different Network Scenarios

Scenario 1 – Public Internet VPN

For a VPN connecting two offices over the public Internet:



ESP is recommended because Internet traffic should normally have confidentiality as well as integrity and authentication.

Scenario 2 – Highly Sensitive Financial Data

Financial information requires confidentiality and integrity.

Examples:

- Banking transactions
- Payment information
- Financial reports
- Customer information

ESP is more appropriate.

Scenario 3 – Non-Confidential but Authenticated Traffic

If an organization has a specific requirement where encryption is unnecessary but integrity and authentication are required, AH may be considered.

However, the deployment must account for AH's limitations and compatibility requirements.

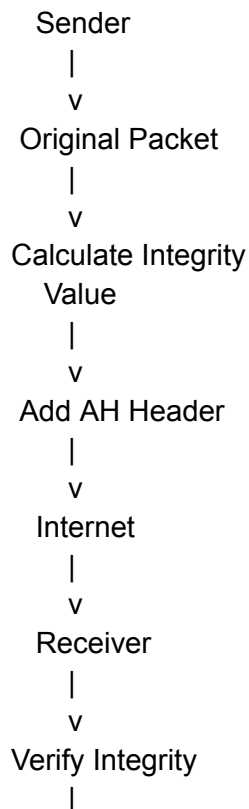
Scenario 4 – Modern Site-to-Site VPN

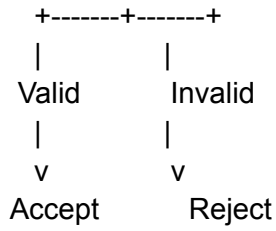
For modern site-to-site VPN deployments:

ESP in tunnel mode is generally preferred.

It can protect the original IP packet and provide confidentiality, integrity, authentication, and replay protection.

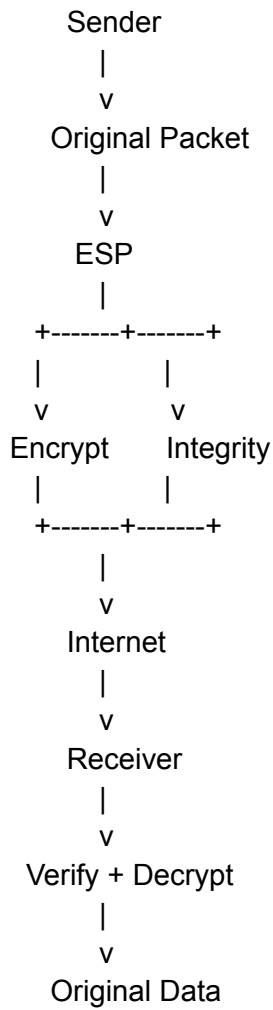
8. AH Packet Flow





The payload is not encrypted.

9. ESP Packet Flow



The protected payload is encrypted before transmission.

10. Advantages of AH

1. Provides data integrity.
 2. Provides data-origin authentication.
 3. Provides anti-replay protection.
 4. Does not require encryption processing.
 5. Can be useful when confidentiality is not required.
-

11. Disadvantages of AH

1. Does not provide confidentiality.
 2. Cannot protect sensitive data from being read.
 3. NAT compatibility is problematic.
 4. Less suitable for modern Internet VPN deployments.
 5. Provides less functionality than ESP for typical VPN requirements.
-

12. Advantages of ESP

1. Provides confidentiality.
 2. Provides integrity.
 3. Supports data-origin authentication.
 4. Supports anti-replay protection.
 5. Suitable for VPNs.
 6. Can protect sensitive business information.
 7. Works well with tunnel-mode site-to-site VPNs.
 8. Supports modern authenticated-encryption configurations.
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13. Disadvantages of ESP

1. Encryption introduces additional processing.

2. Adds packet overhead.
 3. Requires careful key management.
 4. Incorrect configuration can reduce security.
 5. Encryption can increase resource requirements for high-throughput systems.
-

14. Critical Comparison

Criteria	AH	ESP
Integrity	✓	✓
Authentication	✓	✓
Confidentiality	✗	✓
Encryption	✗	✓
Anti-replay	✓	✓
Processing overhead	Lower	Higher
NAT traversal	Poor	Better with NAT-T
Sensitive information	Not sufficient	Appropriate
Site-to-site VPN	Limited	Recommended
Remote-access VPN	Limited	Recommended
Modern deployment	Less common	Common

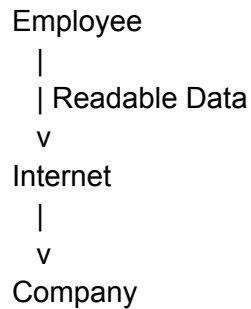
15. Which Is More Appropriate for VPN Deployment?

ESP is more appropriate for modern VPN deployment.

The main reason is that VPNs generally need to protect both the **confidentiality and integrity** of communication.

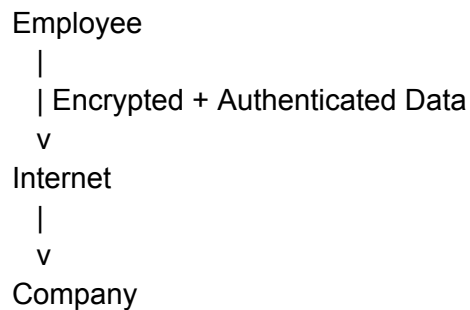
Consider an employee accessing a company server:

Without Confidentiality:



An attacker monitoring the network may be able to observe the content.

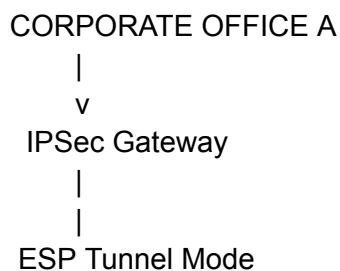
With ESP:

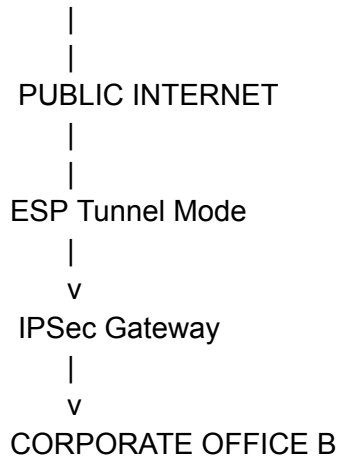


The content is protected from ordinary network interception, while integrity and authentication mechanisms help detect unauthorized modification.

16. Recommended VPN Architecture

For a corporate site-to-site VPN:





Recommended security configuration:

Component	Recommendation
IPSec Protocol	ESP
Mode	Tunnel mode
Key Management	IKEv2
Encryption	AES-based encryption
Integrity	Authenticated encryption or HMAC-based integrity as appropriate
Anti-replay	Enabled
NAT environment	NAT Traversal where required
Key renewal	Automated

17. Security vs Performance Trade-Off

The selection between AH and ESP involves a trade-off.

AH

Lower overhead

+

Integrity

+

Authentication

-

No confidentiality

ESP

Higher overhead

+

Confidentiality

+

Integrity

+

Authentication

+

Anti-replay

For modern enterprise VPNs, the additional processing required by ESP is generally justified by the security benefits.

Hardware acceleration can further reduce the performance impact of encryption.

18. Final Recommendation

For a **modern corporate VPN**, ESP should be selected over AH in most cases.

Recommended configuration:

ESP + Tunnel Mode + IKEv2 + modern authenticated encryption/integrity protection

This provides a strong combination of:

- Confidentiality
- Integrity
- Authentication
- Anti-replay protection
- Secure branch-to-branch communication

AH should only be considered when there is a specific requirement for authentication and integrity without confidentiality and when its networking limitations are acceptable.

19. Conclusion

AH and ESP are important components of IPSec, but they serve different security requirements.

AH provides integrity, authentication, and anti-replay protection but does not encrypt data. It therefore offers limited protection for sensitive communication.

ESP provides confidentiality in addition to integrity, authentication, and anti-replay protection. Although ESP introduces additional processing and packet overhead, modern hardware and optimized cryptographic implementations make it highly suitable for VPNs.

Therefore, **ESP is the preferred IPSec protocol for most modern VPN deployments**, particularly for site-to-site and remote-access VPNs where sensitive information must be protected over untrusted networks.